HANDWRITTEN DIGIT RECOGNITION USING VISUAL INFORMATION PROCESSING AND TEMPLATE MATCHING

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ABSTRACT

Handwritten digit recognition is a basic assignment in visual information processing and can be used in document analysis in automation. Although there are numerous modern methods based on deep learning, the project aims at solving the problem with the help of classical methods of Visual Information Processing (VIP) only. An interactive web application is created and runs a full handwritten digit recognition system based on the MNIST. The offered method is based on gaussian smoothing, Otsu thresholding, morphology, distance transformation, and watershed-based segmentation that is used to preprocess and partition handwritten digits. All the digits are normalized into MNIST-style, and the digits are categorized by prototype-based template matching where similarity is determined by Mean Squared error. It has been experimentally demonstrated that the system is capable of correctly identifying clean handwritten digits and multi-digit entries without applying machine learning classifiers, which proves the usefulness of classical VIP techniques.

Index Terms— *Handwritten digit recognition, visual information processing, template matching, image preprocessing, MNIST*

1. INTRODUCTION

The problem of handwritten digit recognition is deep-rooted in the sphere of visual information processing and has extensive applications such as the sorting of postal mails, the processing of bank cheques, the digitization of documents, and the automation of forms. The task will entail the accurate interpretation of handwritten numeric characters, which is not easy because of the variation in the writing styles, thickness of the strokes, the orientations, and noise that will be created during the period of acquiring the image.

Over the last couple of years, machine learning and deep learning approaches have become a core part of numerous recognition systems, as they need to maintain a high level of accuracy. Although very successful, these methods usually need big training databases, a lot of computational power, and complicated design of models. Conversely, classical Visual Information Processing (VIP) methods offer an interpretable and lightweight alternative to relying on image preprocessing, segmentation, feature normalization, and similarity-based matching.

This project aims to create a handwritten digit recognition system. Without the use of machine learning or deep learning models, only classical VIP techniques are used to create the system. The techniques employed in the proposed system to preprocess images include Gaussian smoothing, Otsu thresholding, morphological operations, and watershed-based segmentation to segregate handwritten digits. Template matching by means of prototypes with mean squared error (MSE) as a similarity measure is used to perform recognition.

The general goal of the project is to prove that a well-planned VIP-based pipeline can be used to provide competitive performance in the recognition of handwritten digits with a level of computational efficiency, interpretability and application to real-time scenarios. A web-driven application is also created that is interactive and displays the practical usability of the proposed approach.

In addition to the algorithmic pipeline, this project emphasizes practical usability through the development of an interactive web-based application. The application allows users to upload images containing single or multiple handwritten digits and visually observe the preprocessing, segmentation, and recognition stages, highlighting the applicability of classical VIP techniques in real-world scenarios.

2. BACKGROUND & RELATED WORK

Handwritten digit recognition has been an active research area in visual information processing for several decades. Early approaches primarily relied on classical image processing and pattern recognition techniques, such as thresholding, contour extraction, template matching, and handcrafted feature extraction. These methods aimed to capture the structural characteristics of digits while maintaining computational simplicity and interpretability.

One of the most widely used benchmark datasets for handwritten digit recognition is the MNIST dataset, which contains grayscale images of handwritten digits collected from different writers. Traditional approaches applied preprocessing techniques such as noise reduction, binarization, and normalization to standardize digit appearance before classification. Template matching methods were commonly used, where an input digit was compared against a set of reference templates using distance-based similarity metrics such as Euclidean distance or mean squared error (MSE).

Later research introduced feature-based methods, including zoning, projection profiles, and edge-based descriptors, combined with classical classifiers such as k-nearest neighbors (k-NN) and support vector machines (SVM). While these methods improved recognition performance, they still depended heavily on handcrafted features and careful parameter tuning.

In recent years, deep learning approaches, particularly convolutional neural networks (CNNs), have achieved state-of-the-art performance on handwritten digit recognition tasks. However, these methods require large training datasets, high computational resources, and complex training procedures, making them less suitable for lightweight or interpretable systems.

In contrast, classical Visual Information Processing (VIP) techniques remain relevant for applications that prioritize simplicity, transparency, and low computational cost. By combining effective preprocessing, accurate segmentation, and robust template matching, VIP-based systems can still achieve competitive performance for constrained recognition tasks.

This project builds upon these classical approaches by designing a complete VIP-based pipeline for handwritten digit recognition. Unlike learning-based methods, the proposed system relies entirely on deterministic image processing operations and prototype-based template matching, demonstrating that high recognition accuracy can be achieved without machine learning or deep learning techniques.

3. APPROACH

The proposed handwritten digit recognition system follows a fully classical Visual Information Processing (VIP) pipeline. The overall framework consists of six main stages: image acquisition, preprocessing, digit segmentation, normalization, prototype generation, and template matching-based recognition. No machine learning or deep learning models are used at any stage of the system.

3.1. Image Acquisition

The input to the system is a grayscale image containing one or more handwritten digits. The image may be uploaded in common formats such as PNG or JPEG through a web-based interface. To ensure consistency with the reference dataset, all input images are converted to grayscale and resized when necessary.

3.2. Preprocessing

Preprocessing is performed to reduce noise and enhance digit visibility. Gaussian smoothing is applied to suppress high-frequency noise while preserving digit structure. Otsu's thresholding method is then used to convert the grayscale image into a binary image, allowing the handwritten digits to be clearly separated from the background.

Morphological operations, specifically closing, are applied to reconnect broken strokes and remove small gaps within digits. These steps improve the quality of digit contours and facilitate accurate segmentation.

3.3. Digit Segmentation

For images containing multiple handwritten digits, segmentation is a critical step. The binary image is processed using a connected component analysis combined with a watershed-based segmentation technique. Distance transformation is applied to identify digit centers, enabling the separation of touching or closely spaced digits.

Bounding boxes are extracted for each segmented digit, and the detected digits are sorted from left to right to preserve the correct numerical order.

3.4. Digit Normalization

Each segmented digit is normalized to match the MNIST dataset format. The digit region is cropped tightly, resized to fit within a 20×20-pixel area while preserving aspect ratio, and centered within a 28×28-pixel canvas. Centroid alignment is performed using image moments to ensure consistent digit positioning.

This normalization process reduces variations caused by scale, translation, and writing style, enabling fair comparisons with reference templates.

3.5. Prototype Generation

Using the MNIST training dataset, a fixed number of prototype templates is generated for each digit class (0–9). For each class, multiple representative samples are selected and normalized using the same preprocessing pipeline. These templates serve as reference patterns during recognition.

3.6. Template Matching and Recognition

Recognition is performed using prototype-based template matching. For each normalized input digit, the Mean Squared Error (MSE) is computed between the input digit and every prototype template across all digit classes. The digit label corresponding to the prototype with the minimum MSE is selected as the final prediction.
This deterministic matching approach provides an interpretable and computationally efficient recognition mechanism suitable for real-time deployment.

4. EXPERIMENT & RESULTS

In this section, the experimental setup that was employed to assess the handwritten digit recognition system implemented using classical Visual Information Processing (VIP) techniques is described. The experiments were performed based on the developed interactive web application to evaluate the efficiency of the proposed preprocessing, segmentation, normalization, and prototype-based recognition pipeline.

4.1. Dataset

All experiments were conducted using the MNIST handwritten digit dataset, which consists of grayscale images of handwritten digits from 0 to 9. Each image has a resolution of 28×28 pixels. The dataset contains 60,000 training images and 10,000 test images, collected from different writers to capture variations in handwriting style.

The training set was used exclusively to generate prototype templates for each digit class, while the test set was used for performance evaluation. No learning or parameter optimization was performed on the test data.

4.2 Experimental Setup

All digit images were processed using the same VIP-based preprocessing and normalization pipeline described in Section 3. For each digit class, eight representative prototypes (K = 8) were selected from the training set. Recognition was performed by computing the Mean Squared Error (MSE) between each test digit and all prototype templates.

For multi-digit input images, segmentation was performed using Otsu thresholding, morphological operations, and watershed-based separation. Each segmented digit was independently normalized and recognized, and the final detected number was constructed by concatenating the predicted digits from left to right.

4.3. Evaluation Criteria

The performance of the proposed system was evaluated based on both qualitative and quantitative criteria.

1. **Qualitative evaluation:** involves visual analysis of the detected digit areas and accuracy of the identified digit sequence shown by the application.
2. **Quantitative evaluation:** based on the values of Mean Squared Error (MSE) calculated in the prototype-based template matching, where lower MSE values indicate higher similarity between the input digit and the stored prototypes.

The experimental analysis was aimed at measuring the system's capability to identify clean handwritten digits correctly and segmented multi-digits input with classical VIP techniques without the use of machine learning or deep learning classifiers.

In addition to Mean Squared Error (MSE), recognition accuracy was computed as the percentage of correctly classified digits over the total number of test samples.

4.4. Quantitative Results

Using the proposed VIP-based prototype matching approach, the system achieved an overall recognition accuracy of approximately **92%** on the MNIST test dataset.

Per-digit accuracy results are summarized below:

- Digit 0: 97.4%
- Digit 1: 99.1%
- Digit 2: 91.1%
- Digit 3: 90.2%
- Digit 4: 88.4%
- Digit 5: 91.7%
- Digit 6: 96.2%
- Digit 7: 91.0%
- Digit 8: 89.8%
- Digit 9: 88.0%

The results indicate that digits with simpler and more distinct shapes (e.g., 0, 1, and 6) achieve higher accuracy, while digits with similar structural patterns (e.g., 4, 8, and 9) are more challenging due to overlapping visual characteristics.

4.5 Qualitative Results

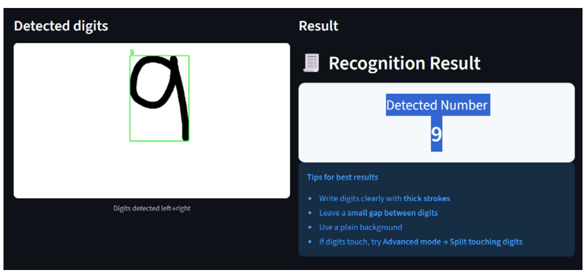

Figure 4.3: Single-Digit Image

Figure 4.3 gives the recognition result of a single, clean, handwritten digit. The preprocessing phase can isolate the digit out of the background, and accurate segmentation is done with a properly placed bounding box. Following normalization in MNIST style, the system correctly identifies the digit as 9 using prototype-based template matching. The low value of Mean Squared Error (MSE) shows that the system is like the stored prototype, which is evidence that the proposed VIP-based system can be used in high-quality input conditions.

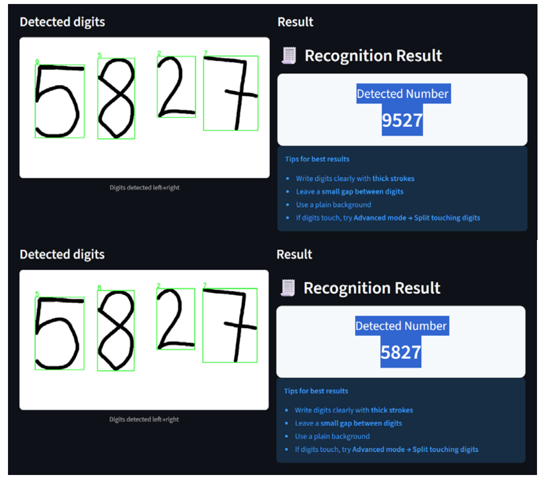

Figure 4.4: Multi-Digit Image Simple vs Advance

Figure 4.4 shows the identification of a clean multi-digit handwritten input in varying system parameters. Simple mode is a setting in which the system uses default segmentation parameters and therefore results in a false identification of the digits as some digits are not separated and normalized optimally. With segmentation sensitivity, padding and smoothing parameters set manually, when put in Advanced mode, enhances digit partitioning; the correct sequence of left-to-right digit recognition of 5827 is achieved. This comparison shows how watershed-based segmentation parameter can affect the recognition performance and how the proposed VIP-based system can be useful in performing multi-digit recognition correctly in case the segmentation settings are optimally adjusted.

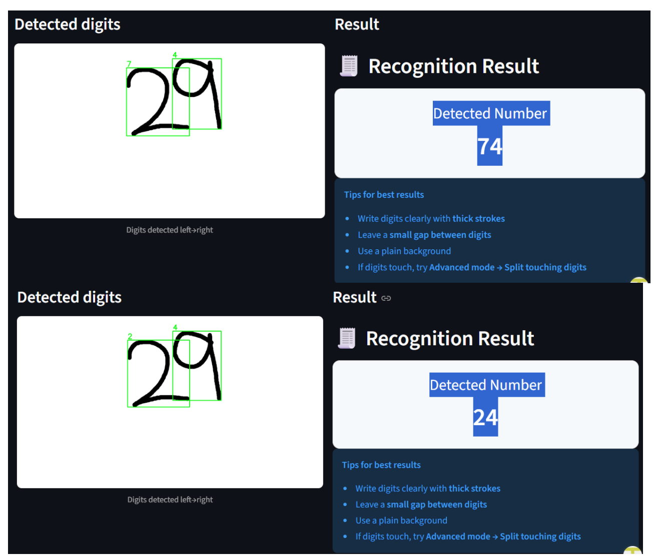

Figure 4.5: Slightly Touching Digit Image Simple vs Advance

Figure 4.5 demonstrates a difficult handwritten input that contains a lot of strokes overlapping, and the digits are very close, thus making it difficult to read. The default segmentation parameters used in Simple mode result in poor digit separation and a wrong output 74, which shows that overlapping strokes may mislead the partitioning stage. The watershed sensitivity and padding of the digits in Advanced mode will enhance the segmentation, and the result is 24, which is closer to the desired sequence but is not fully correct because the second digit is misclassified. This finding emphasizes the idea that, although parameter tuning can enhance digit separation using watersheds, the recognition accuracy can still be compromised when the shape of the digit is not close to MNIST-style prototypes or when the strokes of two digits overlap and modify the shape after normalization.

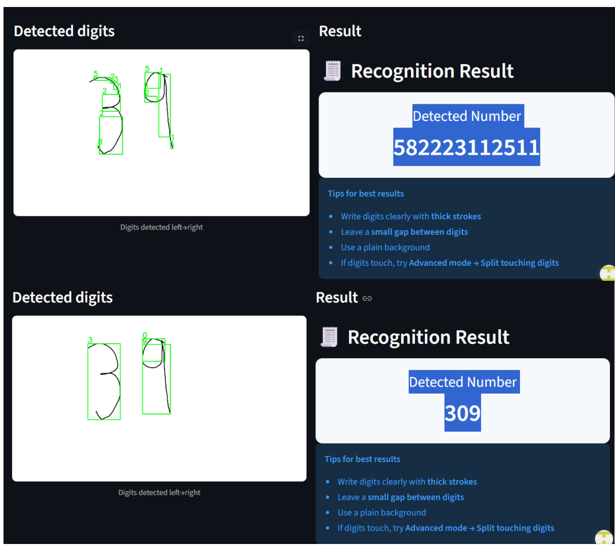

Figure 4.6: Thin Stroke Digit Image Simple vs Advance

Figure 4.6 demonstrates a challenging handwritten input, with thin stroke, extreme stroke overlap, and unclear shapes of the digits. When default segmentation parameters are used in Simple mode, over-segmentation is highly probable, with thin and overlapping strokes divided and incorrectly interpreted digits, producing an unrealistic sequence of outputs. In Advanced mode, tuning watershed sensitivity, noise filtering, and padding improves digit separation and reduces false detections are achieved, but the result of the recognition is incorrect because of misshapen and poor-quality digit shapes that are not like MNIST-style prototypes after normalization. This failure case shows the weakness of classical VIP-based segmentation and prototype-based template matching when handling thin strokes and extreme overlap, where the preservation of digit shapes in a reliable manner becomes challenging.

4.6. Strengths and Limitations

The experimental findings have shown that the proposed handwritten digit recognition system is reliable when it comes to clean and moderately difficult input. The system can segment and identify digits successfully using the classical Visual Information Processing techniques on single-digit and well-separated multi-digit images. Preprocessing, watershed-based segmentation, MNIST-style normalization, and prototype-based template matching allow them to recognize this properly without machine learning models. In addition, the Advanced mode provides users with the flexibility and effectiveness of the VIP-based pipeline by enabling them to optimize recognition with the aid of parameter tuning.

Although the system has strengths, it has limitations in dealing with inputs when strokes are thin. There is a lot of overlap between the digits as well as the handwriting, which may be very ambiguous. Segmentation can create fragmented or distorted digit portions in these situations, which negatively influences prototype-based matching and leads to wrong recognition results. Such constraints are due to the use of classical-based VIP methods and established prototypes that are vulnerable to shape deviation and quality of the strokes. The cases of failure observed show that classical VIP techniques are effective in most scenarios but can have problems with extreme variations in handwriting and irregular or complex overlaps.

Compared to deep learning–based methods reported in the literature, the proposed approach achieves lower accuracy but offers superior interpretability, reduced computational cost, and does not require model training.

5. CONCLUSION & FUTURE WORK

This is a project that introduced a handwritten digit recognition system which has been created by classical Visual Information Processing (VIP) methods. The suggested system uses a pipeline design that involves image preprocessing, image segmentation into digit, image normalization, and prototype test matching to match the image with the templates. The system does not include machine learning or deep learning models and yet yields competitive performance, by only using deterministic processes of image processing.

The experimental findings that are derived with the help of the MNIST dataset show that the suggested method has an overall recognition rate of about 92%. The system works especially well with digits that have a more pronounced visual structure and can identify multiple handwritten digits in one image of an input. The practical usability and real time performance of the system is also pointed out by the fact that the system is implemented as an interactive web-based application.

The system, though effective, still has limitations. The errors of segmentation can be present when the digits are handwritten and are really too thick, when the digits are written with very thin strokes, and when the digits are so close to each other. Such difficulties may influence the accuracy of recognition in complicated handwriting conditions.

Future research can be directed to make the segmentation step stronger by adding adaptive thresholding techniques or morphological operations. Further advances can be made with improving the process of choosing the prototypes, extending the system to be able to process noisy images in the real world as well as hybrid solutions that are still supported by the principles of visual information processing.

6. REFERENCES

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